

Shrey Tiwari

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My research focuses on ensuring software reliability through a diverse set of approaches, including building precise static and dynamic analysis tools, developing automated testing techniques, and more recently, leveraging software agents to detect and prevent bugs at scale.

Education

Carnegie Mellon University - School of Computer Science

Doctor of Philosophy, Advisor: Prof. Rohan Padhye

Aug 2023 - Jul 2028 (expected)

CGPA: **4.0/4.0**

PES University

B.Tech (honors) in Computer Science & Engineering (Systems & Core Computing), **Gold Medalist**

Aug 2016 - Jul 2020

CGPA: **9.93/10.0**

Research

Chronically Buggy: Analyzing Date/Time Pitfalls In Open-Source And LLM-Generated Python Code

EMSE 2025

Shrey Tiwari, Alexander Joukov, Peter Vandervelde, Serena Chen, Leyi Cui, Ao Li, Rohan Padhye

Under Review

It's About Time: An Empirical Study of Date and Time Bugs in Open-Source Python Software

MSR 2025

Shrey Tiwari, Alexander Joukov, Peter Vandervelde, Serena Chen, Ao Li, Rohan Padhye

🏆 ACM SIGSOFT Distinguished Paper Award

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM

OOPSLA 2025

Ao Li, Byeongjee Kang, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye

Lightweight and Modular Resource Leak Checking (Extended Version)

SOAP 2024

Pritam Gharat, Narges Shadab, Shrey Tiwari, Michael D. Ernst,... Shuvendu Lahiri, Akash Lal, Manu Sridharan

Inference of Resource Management Specifications

OOPSLA 2023

Pritam Gharat, Narges Shadab, Shrey Tiwari, Michael D. Ernst,... Shuvendu Lahiri, Akash Lal, Manu Sridharan

Resource Leak Checker (RLC#) for C# code using CodeQL

Report 2023

Narges Shadab, Pritam Gharat, Shrey Tiwari, Shuvendu Lahiri, Akash Lal

Awards & Honors

Hima and Jive Fellowship

Nov 2025

Recognized for research focusing on improving the reliability of date/time computations in software

Distinguished Paper Award

Apr 2025

Awarded the ACM SIGSOFT Distinguished Paper Award at MSR 2025 in Ottawa

SIGPLAN Travel Support

Oct 2024

Granted travel support to attend SPLASH 2024 in Pasadena

Gold Medalist

May 2020

Awarded a gold medal for ranking second in the entire batch of 500+ CSE students, PES University

CNR Rao Merit Scholarship

2017 - 2020

Six-time recipient of the CNR Rao Merit Scholarship (every applicable semester)

MHRD Scholarship

2017 - 2020

Six-time recipient of the MHRD Merit Scholarship (every applicable semester)

Work Experience

Uber Technologies, Inc.

Jun 2025 - Aug 2025

Research Intern | Advisors: Dr. Sonal Mahajan [✉](#) & Dr. Yuxin Wang [✉](#)

Sunnyvale, CA

- Worked with the Programming Systems group to improve software quality in production by improving automated program analysis.
- Enhanced the accuracy of automated code reviews in uReview [✉](#) and improved the precision of the static nil-panic detector, NilAway [✉](#).

Microsoft Research

Jun 2021 - Jul 2023

Research Fellow | Advisors: Dr. Akash Lal [✉](#) & Dr. Suman Nath [✉](#)

Bangalore, India

- At Microsoft Research India's Cloud Reliability team, I developed tools to "shift left" Azure reliability and prevent production bugs. My research focused on developing static and dynamic resource leak detection tools for the Microsoft Azure cloud.

Citrix Systems

Jul 2020 - Jun 2021

Software Engineer | Supervisor: Mukul Agarwal [✉](#)

Bangalore, India

- I was the core developer for the Citrix Cloud Connector [✉](#), a critical component of the cloud VPN gateway service, handling over 3 million weekly VPN tunnels.
- I was responsible for feature development, bug fixes, production monitoring, and on-call support. I contributed to increasing the test coverage to >90%, resulting in the Cloud Connector becoming the most reliable component of the cloud VPN solution.

Citrix Systems

Jan 2020 - Jun 2020

Software Development Intern | Supervisor: Mukul Agarwal [✉](#)

Bangalore, India

- Developed a test automation platform for Citrix VPN Gateway client, eliminating manual testing efforts and reducing testing time from 2 days to 5 hours.

Morgan Stanley

May 2019 - Jul 2019

Technology Analyst Intern | Supervisor: Vignesh Natarajan [✉](#)

Bangalore, India

- I containerized the "Data Influx Adapter", a key component of the tax lot processing pipeline, using Docker. Testing on a 6-million tax lot dataset demonstrated significant cost savings while maintaining performance. I successfully deployed the containerized solution to production, improving code portability and developer productivity.

Indian Institute of Science

May 2018 - Aug 2018

Research Intern | Advisor: Prof. Arkaprava Basu [✉](#)

Bangalore, India

- I was selected as one of 19 students (from 2500+ applicants) for the Narendra Summer Internship [✉](#) at IISc's Computer Systems Lab [✉](#), researching dynamic detection of heterogeneous data races in CUDA-based GPGPU programs.

Selected Projects

DateSMT: SMT-Based Reasoning For Calendrical Arithmetic

Sept 2025 - Present

Carnegie Mellon University

- Building DateSMT, the first framework to encode real-world date and period semantics into SMT solvers, unlocking more precise and scalable analyses for tools that rely on symbolic execution, model checking, and formal verification.

AI-Assisted Differential Fuzz Testing Of Date/Time Libraries

Mar 2025 - Present

Carnegie Mellon University

- Developing a differential fuzz testing framework that utilizes LLMs to synthesize and translate date/time computations across libraries, aimed at detecting edge-case bugs and ensuring the correctness of Python date/time libraries.

NilAway: A Practical Nil Panic Detection Tool for Go

Jul 2025 - Dec 2025

Uber Technologies, Inc.

- Implemented an agentic-AI-driven post-processing phase for NilAway that improves precision by reasoning over reported alerts, automatically synthesizing unit tests to validate nil panic bugs, and prioritizing findings with reproducible failures.

FixIt: A Precise And Scalable Dynamic Resource-Leak Detection Tool

Aug 2021 - Jul 2022

Microsoft Research

- Developed a novel and scalable lightweight instrumentation-based tool for detecting resource leaks in managed programming languages like C# and Java, achieving approximately 79% accuracy by leveraging unit test suites.

Selected Talks, Posters, & Media

Program Analysis In Practice: Lessons From Statically Analyzing Millions Of Lines Of Code [↗](#)

Carnegie Mellon University

Oct 2025

Invited Guest Lecture

The Date & Time Bugs Crashing Your Code [↗](#)

Disseminate: The Computer Science Research Podcast

Sep 2025

Invited Podcast

uReview: Improving the Precision Of Code Reviews At Industry Scale

Uber Technologies, Inc.

Aug 2025

Research Presentation

FixIt: Precise And Scalable Dynamic Detection Of Resource Leaks

Microsoft Research (Cloud Reliability Workshop)

Feb 2022

Invited Research Presentation

Dynamic Detection of Heterogeneous Data Races In CUDA-Based GPGPU Programs

Indian Institute of Science (IISc)

Jul 2018

Research Poster

Mentoring

Research Study

Mentored an undergraduate student on her final year research project at the PASTA Lab, where she collaborated with me on my ongoing research.

Dec 2024 - Apr 2025

REUSE Program

Mentored three outstanding undergraduate students during the summer, contributing to CMU's recognition as having the best REU mentors in the United States.

Jun 2024 - Aug 2024

Service

Student Reviewer

PLDI 2025, ISSTA 2024

Student Volunteer

SPLASH 2024

Steering Committee Member

Software Engineering Seminar [↗](#), 2025–2026